The SU(3) Gram Cone and the Rank-Six Toroidal Lattice

Exact orbit spaces, Weyl-chamber incidence, and the obstruction to a monodromy-compatible Gram map

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Abstract

Let SU(3) act on $\mathrm{Mat}_{3\times 2}(\mathbb{C})$ by left multiplication. The first part proves that the Gram map induces a proper stratified homeomorphism from the orbit space onto the positive-semidefinite 2×2 Hermitian cone, determines every stabilizer and orientation, and identifies the compact quotient as B^4, not S^4. The second part treats the distinct rank-six toroidal lattice $\Lambda_3 = H_1(T^6;\mathbb{Z}) = L_3 \oplus \bar{L}_3$. We compute its exact diagonal S_3-action, integral symplectic monodromy, and the quotient $\Phi = P \oplus Q$. Its kernel is a fixed rank-two lattice, while each rank-two quotient is the A_2 weight lattice rather than the root lattice; the sequence splits as abelian groups but has no integral S_3-equivariant splitting. The six Kuhn chambers form the regular S_3-set, not the two three-point orbits furnished by a basis of Λ_3. An exact incidence complex and the spectral-floor evaluation place the six chambers on six diagonal rank-two Gram-boundary points, but the tempting extension from all six lattice directions has a non-invariant kernel and therefore cannot intertwine monodromy. Finally we prove that the branch-exponent map restricts to the angular map P; composing it with the deck map Q is a domain error. The separate free defect lattice $H^4(U;\mathbb{Z}) \cong (\mathbb{Z}^2)^3$ is not identified with Λ_3, despite the equality of ranks.

Status and scope. This note separates two exact carriers. The rank-two Gram quotient follows from the displayed SU(3) action. The later toroidal calculation uses the explicitly marked lattice Λ_3, its Kuhn fan, and $\Phi = P \oplus Q$; it does not assert a holomorphic map from the toroidal total space to a Gram or Jordan cone. The representation sphere S^{12}_{col} is not a claimed complex manifold and is not automatically a cusp-induced smash sphere. The note constructs neither a nonzero instanton sector on S^4 nor a quantum Yang–Mills Hamiltonian. The rank-three toroidal lattice and the regular-locus defect group are kept as different typed objects throughout.

Provenance. This standalone draft extracts the rigorously surviving Gram-quotient and toroidal-lattice calculations from independent audits of proposed continuations of the reconstructed S^6 manuscript. The orbit, stabilizer, orientation, compactification, clutching, lattice, incidence, and obstruction claims are proved below from their stated objects; no theorem about the existence of the candidate complex structure on S^6 is used.

1 The higher-rung SU(3) Gram cone

This section isolates the part of the proposed higher-rung construction that is independent of the candidate complex threefold and proves it from the underlying linear action. The compact

object used here is the representation sphere

$$\mathrm{Mat}_{3\times 2}(\mathbb{C})^{+},$$

the one-point compactification of a twelve-real-dimensional vector space. It must not be identified with a cusp-induced S^{12} construction unless an additional based identification has first been chosen. The proofs below also distinguish three structures that otherwise look deceptively similar: the Lorentzian determinant on $\mathrm{Herm}_2(\mathbb{C})$, the positive-semidefinite Gram cone inside that vector space, and a separately chosen Euclidean metric on the positive-definite cone.

1.1 Conventions and the Gram morphism

Put

$$\mathscr{V} = \mathrm{Mat}_{3\times 2}(\mathbb{C}), \qquad \Gamma : \mathscr{V} \longrightarrow \mathrm{Herm}_2(\mathbb{C}), \qquad \Gamma(A) = A^{\dagger}A. \tag{1.1}$$

Write $A = (a\ b)$, with $a, b \in \mathbb{C}^3$, and let $\mathrm{SU}(3)$ act on the left:

$$U \cdot A = UA.$$

For $r = 0, 1, 2$, let

$$\mathscr{V}_{(r)} = \{A \in \mathscr{V} : \mathrm{rank}_{\mathbb{C}}\, A = r\}.$$

The Hermitian products are linear in the second argument. We orient $\mathscr{V}$ by the complex orientation, in the real coordinate order

$$\begin{aligned}(&\mathrm{Re}\, A_{11}, \mathrm{Im}\, A_{11}, \mathrm{Re}\, A_{12}, \mathrm{Im}\, A_{12}, \mathrm{Re}\, A_{21}, \mathrm{Im}\, A_{21},\\ &\mathrm{Re}\, A_{22}, \mathrm{Im}\, A_{22}, \mathrm{Re}\, A_{31}, \mathrm{Im}\, A_{31}, \mathrm{Re}\, A_{32}, \mathrm{Im}\, A_{32}).\end{aligned} \tag{1.2}$$

On $\mathrm{Herm}_2(\mathbb{C})$, use the oriented real-linear coordinates

$$\Phi(t, x, y, z) = \begin{pmatrix} t+z & x-iy \\ x+iy & t-z \end{pmatrix}, \qquad \mathrm{or}_{\mathrm{Herm}} = dt \wedge dx \wedge dy \wedge dz. \tag{1.3}$$

Thus, if $\Gamma(A)_{12} = \langle a, b\rangle$, then

$$\begin{aligned} t &= \frac{1}{2}(\|a\|^2 + \|b\|^2), & z &= \frac{1}{2}(\|a\|^2 - \|b\|^2),\\ x &= \mathrm{Re}\langle a, b\rangle, & y &= -\,\mathrm{Im}\langle a, b\rangle. \end{aligned} \tag{1.4}$$

1.2 Exact orbit classification

Theorem 1.1 (The Gram quotient)**.** *The morphism* Γ *is* $\mathrm{SU}(3)$*-invariant, has image exactly* $\mathrm{Herm}_2(\mathbb{C})_{\geq 0}$*, and separates* $\mathrm{SU}(3)$*-orbits. It induces a proper homeomorphism*

$$\overline{\Gamma} : \mathscr{V}/\mathrm{SU}(3) \xrightarrow{\cong} \mathrm{Herm}_2(\mathbb{C})_{\geq 0}. \tag{1.5}$$

Proof. For $U \in \mathrm{SU}(3)$,

$$\Gamma(UA) = A^{\dagger}U^{\dagger}UA = A^{\dagger}A,$$

so Γ is invariant. Every matrix $A^{\dagger}A$ is positive semidefinite. Conversely, for $H \in \mathrm{Herm}_2(\mathbb{C})_{\geq 0}$, let $H^{1/2}$ be its unique positive-semidefinite square root and define

$$s(H) = \begin{pmatrix} H^{1/2} \\ 0 \quad 0 \end{pmatrix} \in \mathrm{Mat}_{3\times 2}(\mathbb{C}). \tag{1.6}$$

Then $s(H)^\dagger s(H) = H$, proving surjectivity and giving a continuous section.

It remains to prove exact orbit separation. Suppose $A^\dagger A = B^\dagger B$. Define

$$T : \operatorname{im} A \longrightarrow \operatorname{im} B, \qquad T(Au) = Bu.$$

If $Au = 0$, then

$$0 = \|Au\|^2 = u^\dagger A^\dagger A u = u^\dagger B^\dagger B u = \|Bu\|^2,$$

so $Bu = 0$, and T is well defined. For $u, v \in \mathbb{C}^2$,

$$\langle T(Au), T(Av)\rangle = u^\dagger B^\dagger B v = u^\dagger A^\dagger A v = \langle Au, Av\rangle.$$

Thus T is a unitary isomorphism of the image subspaces. Extend it to some $U_0 \in \mathrm{U}(3)$ satisfying $U_0 A = B$. Since $\operatorname{rank} B \le 2$, the orthogonal complement of $\operatorname{im} B$ has positive complex dimension. Choose a unitary map V which fixes $\operatorname{im} B$ pointwise and whose determinant is $(\det U_0)^{-1}$. Then

$$U = V U_0 \in \mathrm{SU}(3), \qquad UA = VB = B.$$

Hence equal Gram matrices are exactly equal SU(3)-orbits.

The action is proper because SU(3) is compact; hence the orbit space is Hausdorff. The continuous map $H \mapsto [s(H)]$ is inverse to the continuous induced map $\overline{\Gamma}$, so (1.5) is a homeomorphism. Finally,

$$\|A\|_F^2 = \operatorname{tr}(A^\dagger A) = \operatorname{tr} \Gamma(A). \tag{1.7}$$

If K is compact in the positive-semidefinite cone, then $\Gamma^{-1}(K)$ is closed and bounded by (1.7), hence compact. Thus Γ is proper. For compact K, $\overline{\Gamma}^{-1}(K)$ is the continuous image in the orbit space of the compact set $\Gamma^{-1}(K)$, so $\overline{\Gamma}$ is proper as well. □

Proposition 1.2 (Stabilizers and orbit-type strata)**.** *For $A \in \mathscr{V}_{(r)}$,*

$$\operatorname{Stab}_{\mathrm{SU}(3)}(A) \cong \mathrm{SU}(3-r). \tag{1.8}$$

The exact dimensions are

r	$\dim_{\mathbb{R}} \mathscr{V}_{(r)}$	$\operatorname{Stab}(A)$	$\dim_{\mathbb{R}}(\mathrm{SU}(3)\cdot A)$	$\dim_{\mathbb{R}}(\mathscr{V}_{(r)}/\mathrm{SU}(3))$
2	12	$\{1\}$	8	4
1	8	SU(2)	5	3
0	0	SU(3)	0	0.

(1.9)

Consequently the quotient has topological dimension four, but the action is free only on the rank-two stratum.

Proof. The equality $UA = A$ is equivalent to the assertion that U fixes $\operatorname{im} A$ pointwise. Restriction to its orthogonal complement identifies the stabilizer with $\mathrm{SU}(3-r)$, proving (1.8). The rank-two locus is open in the twelve-real-dimensional vector space. The complex rank-one locus in the space of 3×2 matrices has complex dimension

$$1(3 + 2 - 1) = 4,$$

hence real dimension eight away from the origin. Subtracting $8 - \dim_{\mathbb{R}} \operatorname{Stab}(A)$ gives the displayed orbit and quotient dimensions. They agree with the target strata: positive-definite Hermitian matrices form a four-dimensional open cone, while every nonzero rank-one positive-semidefinite matrix has a unique expression $\lambda v v^\dagger$ with $\lambda > 0$ and $v \in S^3$, modulo $v \sim e^{i\theta} v$. Hence that target stratum is

$$(0, \infty) \times \mathbb{CP}^1 \cong (0, \infty) \times S^2,$$

of real dimension three. The rank-zero stratum is the origin. □

1.3 The free locus, its explicit trivialization, and orientation

Theorem 1.3 (Determinant-one completion)**.** *There is an* SU(3)*-equivariant diffeomorphism*

$$\mathscr{V}_{(2)} \xrightarrow{\cong} \mathrm{SU}(3) \times \mathrm{Herm}_2(\mathbb{C})_{>0} \tag{1.10}$$

under which Γ *is projection to the second factor and the group acts by left multiplication on the first. Thus the principal bundle over the free quotient has a global section and is trivial.*

Proof. For $A \in \mathscr{V}_{(2)}$, set

$$H = A^\dagger A > 0, \qquad Q = AH^{-1/2} = (q_1\ q_2).$$

Then $Q^\dagger Q = I_2$, so (q_1, q_2) is an ordered orthonormal two-frame in $\mathbb{C}^3$. With $\varepsilon_{123} = 1$, define

$$(q_3)_i = \sum_{j,k=1}^{3} \varepsilon_{ijk}\overline{(q_1)_j}\,\overline{(q_2)_k}. \tag{1.11}$$

Antisymmetry gives $q_1^\dagger q_3 = q_2^\dagger q_3 = 0$. The Gram–Plücker identity, proved independently in theorem 1.4, gives $\|q_3\| = 1$. Finally, contraction of the two alternating tensors gives

$$\begin{aligned}\det[q_1\ q_2\ q_3] &= \sum_{i,j,k} \varepsilon_{ijk}(q_1)_i(q_2)_j(q_3)_k \\ &= \|q_1\|^2\|q_2\|^2 - |q_1^\dagger q_2|^2 = 1.\end{aligned}$$

Thus $U_A = [q_1\ q_2\ q_3] \in \mathrm{SU}(3)$, and

$$A = U_A s(H),$$

where s is (1.6). The positive square root, inverse square root, and the polynomial-conjugate formula (1.11) are smooth on the positive-definite locus. Consequently

$$A \longmapsto (U_A, A^\dagger A), \qquad (U, H) \longmapsto Us(H)$$

are smooth inverse maps. Their equivariance is immediate. □

The order of factors in (1.10) does not by itself fix an orientation. We now fix and calculate it. Give $\mathfrak{su}(3)$ the ordered real basis

$$\begin{gathered} X_{12}, Y_{12}, X_{13}, Y_{13}, X_{23}, Y_{23}, D_1, D_2, \\ X_{jk} = E_{jk} - E_{kj}, \qquad Y_{jk} = i(E_{jk} + E_{kj}), \\ D_1 = i\,\mathrm{diag}(1, -1, 0), \qquad D_2 = i\,\mathrm{diag}(0, 1, -1). \end{gathered} \tag{1.12}$$

At

$$A_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix},$$

use the oriented Hermitian basis

$$K_t = I_2, \quad K_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad K_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad K_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{1.13}$$

and horizontal lifts $B_\alpha = \frac{1}{2}A_0K_\alpha$. Directly, $d\Gamma_{A_0}(B_\alpha) = K_\alpha$. Relative to (1.2), the twelve columns

$$(X_{12}A_0, Y_{12}A_0, X_{13}A_0, Y_{13}A_0, X_{23}A_0, Y_{23}A_0, D_1A_0, D_2A_0, B_t, B_x, B_y, B_z)$$

form the exact real coordinate matrix

$$M_{\mathrm{or}} := \begin{pmatrix} 0&0 & 0&0 & 0&0 & 0&0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0&0 & 0&0 & 0&0 & 1&0 & 0 & 0 & 0 & 0 \\ 1&0 & 0&0 & 0&0 & 0&0 & 0 & \frac{1}{2} & 0 & 0 \\ 0&1 & 0&0 & 0&0 & 0&0 & 0 & 0 & -\frac{1}{2} & 0 \\ -1&0 & 0&0 & 0&0 & 0&0 & 0 & \frac{1}{2} & 0 & 0 \\ 0&1 & 0&0 & 0&0 & 0&0 & 0 & 0 & \frac{1}{2} & 0 \\ 0&0 & 0&0 & 0&0 & 0&0 & \frac{1}{2} & 0 & 0 & -\frac{1}{2} \\ 0&0 & 0&0 & 0&0 & -1&1 & 0 & 0 & 0 & 0 \\ 0&0 & -1&0 & 0&0 & 0&0 & 0 & 0 & 0 & 0 \\ 0&0 & 0&1 & 0&0 & 0&0 & 0 & 0 & 0 & 0 \\ 0&0 & 0&0 & -1&0 & 0&0 & 0 & 0 & 0 & 0 \\ 0&0 & 0&0 & 0&1 & 0&0 & 0 & 0 & 0 & 0 \end{pmatrix}. \tag{1.14}$$

Laplace expansion gives

$$\det M_{\mathrm{or}} = \frac{1}{2} > 0. \tag{1.15}$$

Since both factors in (1.10) are connected, the sign is constant on the free locus. Therefore the quotient orientation determined by

$$\mathrm{or}_{\mathscr{V}} = \mathrm{or}_{\mathrm{SU}(3)} \wedge \mathrm{or}_{\mathscr{V}_{(2)}/\mathrm{SU}(3)}$$

agrees under Γ with $dt \wedge dx \wedge dy \wedge dz$.

1.4 Lorentz determinant and Plücker norm

For $H = \Phi(t, x, y, z)$, direct expansion gives

$$\operatorname{tr} H = 2t, \qquad \det H = t^2 - x^2 - y^2 - z^2, \tag{1.16}$$

and the eigenvalues are $t \pm \sqrt{x^2 + y^2 + z^2}$. Hence

$$\begin{aligned} H > 0 &\iff t > \sqrt{x^2 + y^2 + z^2}, \\ H \geq 0 &\iff t \geq \sqrt{x^2 + y^2 + z^2}, \\ \operatorname{rank} H = 1 &\iff t = \sqrt{x^2 + y^2 + z^2} > 0. \end{aligned} \tag{1.17}$$

Thus det has signature $(1, 3)$ when positive directions are listed first, while $-\det$ has signature $(3, 1)$. The quotient image is not all of this Lorentz vector space: it is precisely the closed future causal cone, and its free locus is the open future timelike cone.

Proposition 1.4 (Gram–Plücker identity)**.** *For $A = (a\ b)$,*

$$\det \Gamma(A) = \|a\|^2\|b\|^2 - |\langle a, b\rangle|^2 = \sum_{1 \leq i < j \leq 3} |a_i b_j - a_j b_i|^2 = \|a \wedge b\|^2. \tag{1.18}$$

Proof. Expanding gives

$$\begin{aligned} \|a\|^2\|b\|^2 - |\langle a, b\rangle|^2 &= \sum_{i,j} |a_i|^2 |b_j|^2 - \sum_{i,j} \overline{a_i} b_i a_j \overline{b_j} \\ &= \sum_{i<j} \big(|a_i|^2|b_j|^2 + |a_j|^2|b_i|^2 - \overline{a_i} b_i a_j \overline{b_j} - \overline{a_j} b_j a_i \overline{b_i}\big) \\ &= \sum_{i<j} |a_i b_j - a_j b_i|^2. \end{aligned}$$

The last expression is the squared Hermitian norm of $a \wedge b$ in the ordered basis $(e_1 \wedge e_2, e_1 \wedge e_3, e_2 \wedge e_3)$. □

The determinant is only one invariant and does not classify orbits. For example,

$$A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} \sqrt{2} & 0 \\ 0 & 1/\sqrt{2} \\ 0 & 0 \end{pmatrix}$$

satisfy $\det \Gamma(A_1) = \det \Gamma(A_2) = 1$, but their Gram matrices are I_2 and $\mathrm{diag}(2, 1/2)$, so theorem 1.1 shows that they lie in distinct orbits.

1.5 The four-coefficient form is additional input

Let $r = (r_1, r_2, r_3, r_4)^{\mathsf{T}} \in \mathbb{R}^4$, and define the orthogonal Hadamard coordinates

$$\begin{pmatrix} h \\ x \\ y \\ z \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{pmatrix}. \tag{1.19}$$

The proposed coefficient-space form

$$Q_4(r) = \frac{1}{2} \left(\sum_{i=1}^{4} r_i \right)^2 - \sum_{i=1}^{4} r_i^2 \tag{1.20}$$

satisfies the exact identity

$$Q_4(r) = h^2 - x^2 - y^2 - z^2. \tag{1.21}$$

Indeed, orthogonality gives $\sum r_i^2 = h^2 + x^2 + y^2 + z^2$ and $\sum r_i = 2h$.

The same Hermitian matrix can be written in the alternative convention

$$\Psi(h, x, y, z) = \begin{pmatrix} h + y & x + iz \\ x - iz & h - y \end{pmatrix}. \tag{1.22}$$

It agrees exactly with Φ under

$$(t, x, y_\Phi, z_\Phi) = (h, x, -z, y).$$

The transformation $(y, z) \mapsto (-z, y)$ has determinant $+1$, so the two conventions give the same matrix, Lorentz form, and orientation.

Nothing in the SU(3) orbit calculation selects the coefficient $1/2$. More generally,

$$Q_{4,\lambda}(r) = \lambda \left(\sum_i r_i \right)^2 - \sum_i r_i^2 = (4\lambda - 1)h^2 - x^2 - y^2 - z^2. \tag{1.23}$$

Every $\lambda > 1/4$ therefore has Lorentz signature $(1, 3)$, after a rescaling of h. The value $\lambda = 1/2$ is the unique member for which the fixed orthogonal coordinates in (1.19) have equal unit coefficients. This is a normalization choice in coefficient space, not a theorem forced by the Gram quotient.

Nor does a nonnegative coefficient tuple automatically determine a point of the positive Gram cone. The tuple $(1, 0, 0, 0)$ has $Q_4 = -1/2$. A coefficient tuple enters the Gram quotient through (1.22) only when that Hermitian matrix is positive semidefinite, equivalently when $h \geq 0$ and $Q_4 \geq 0$. The hypersurface $z = 0$ is a real-symmetric coefficient slice, but that fact alone does not supply a coordinatewise nonnegative embedding of a three-component mass domain.

1.6 The compact representation quotient

Theorem 1.5 (The S^{12} representation quotient)**.** *Let*

$$S^{12}_{\mathrm{col}} = \mathscr{V}^{+} = \mathscr{V} \cup \{\infty\}$$

be the one-point compactification, and extend the SU(3) *action by fixing* ∞*. Then*

$$S^{12}_{\mathrm{col}}/\mathrm{SU}(3) \cong \left(\mathrm{Herm}_2(\mathbb{C})_{\geq 0}\right)^{+} \cong B^4. \tag{1.24}$$

The orbit-type strata are

source stratum	*quotient stratum*	*stabilizer*
$\operatorname{rank} A = 2$	$\operatorname{int} B^4$	$\{1\}$
$\operatorname{rank} A = 1$	$\partial B^4 \setminus \{p_0, p_\infty\}$	$\mathrm{SU}(2)$
$A = 0$	$\{p_0\}$	$\mathrm{SU}(3)$
$A = \infty$	$\{p_\infty\}$	$\mathrm{SU}(3)$.

(1.25)

The ball identification is topological on the full stratified quotient. On its smooth interior, it may be chosen to carry the quotient orientation to the orientation fixed in (1.3)*.*

Proof. As a real vector space, $\mathscr{V} \cong \mathbb{R}^{12}$, so its one-point compactification is S^{12}. On $\mathscr{V} \cong \mathbb{C}^3 \oplus \mathbb{C}^3$, the action of $U \in \mathrm{SU}(3)$ has complex determinant $(\det U)^2 = 1$, hence real determinant one; it preserves the standard orientation before and after one-point compactification.

The properness proved from (1.7) implies that Γ extends continuously to

$$\Gamma^{+} : \mathscr{V}^{+} \longrightarrow \left(\mathrm{Herm}_2(\mathbb{C})_{\geq 0}\right)^{+}, \qquad \Gamma^{+}(\infty) = \infty.$$

It separates the finite orbits by theorem 1.1, and the added point is a distinct fixed orbit. Hence it induces a continuous bijection

$$\mathscr{V}^{+}/\mathrm{SU}(3) \longrightarrow \left(\mathrm{Herm}_2(\mathbb{C})_{\geq 0}\right)^{+}.$$

The source is compact Hausdorff because a compact group acts on the compact Hausdorff space $\mathscr{V}^{+}$. The target is Hausdorff because the positive-semidefinite cone is locally compact Hausdorff. The continuous bijection is therefore a homeomorphism.

By (1.17), the map

$$(t, \mathbf{x}) \longmapsto (t - \|\mathbf{x}\|, \mathbf{x})$$

is a homeomorphism from the closed future cone to the closed half-space $\mathbb{R}_{\geq 0} \times \mathbb{R}^3$. For $u = (s, \mathbf{x})$ in that half-space, inverse stereographic projection is

$$u \longmapsto \frac{1}{1 + \|u\|^2}\,(2s, 2\mathbf{x}, \|u\|^2 - 1) \in S^4 \subset \mathbb{R}^5. \tag{1.26}$$

Its image is the closed hemisphere with first coordinate nonnegative, minus the north pole. The point at infinity supplies that pole. A closed hemisphere is a closed four-ball, proving (1.24).

Positive-definite matrices correspond to the open half-space and hence the ball interior. Nonzero rank-one matrices correspond to $\{0\} \times (\mathbb{R}^3 \setminus \{0\})$. The zero matrix and the point at infinity give two distinct endpoints in the compactification of that three-dimensional boundary. Combining this with theorem 1.2 proves (1.25). The orientation statement on the connected interior is the calculation following (1.15). □

The notation S^{12}_{col} records the construction. There is a canonical pointed homeomorphism of representation spheres

$$(\mathbb{C}^3)^+ \wedge (\mathbb{C}^3)^+ \cong (\mathbb{C}^3 \oplus \mathbb{C}^3)^+ = S^{12}_{\mathrm{col}}.$$

If a different based six-sphere X is used, a based homeomorphism $X \cong (\mathbb{C}^3)^+$ transports the action to $X \wedge X$. The transported action depends on that choice. Without an independently proved equivariance statement, it is not induced by a torus fibration, cusp monodromy, period matrix, or fibrewise flux on X.

1.7 Exact gauge-theoretic boundary

Theorem 1.6 (No native nonzero S^4 sector from the Gram quotient)**.** *The construction above supplies no native principal* SU(3) *bundle of nonzero second Chern number on S^4. More precisely:*

1. *the orbit projection over the rank-two locus is the product bundle in (1.10), so its second Chern class is zero;*
2. *on the rank-one boundary the orbit projection is not a principal* SU(3)*-bundle, because its stabilizer is* SU(2)*;*
3. *the full compact quotient is B^4, not S^4; and*
4. *replacing B^4 by the one-point compactification of its interior is a different construction whose clutching class is additional data.*

Proof. The first two assertions are theorems 1.2 and 1.3; the third is theorem 1.5. For the fourth, the matrix logarithm is a diffeomorphism

$$\log : \mathrm{Herm}_2(\mathbb{C})_{>0} \xrightarrow{\cong} \mathrm{Herm}_2(\mathbb{C}) \cong \mathbb{R}^4.$$

Its one-point compactification is therefore S^4, but this operation collapses the entire finite rank-one boundary of the full Gram quotient into the new point at infinity. Principal SU(3)-bundles on this new S^4 are obtained by clutching two trivial bundles over hemispheres with a map

$$\kappa : S^3 \longrightarrow \mathrm{SU}(3).$$

The fibration $\mathrm{SU}(2) \to \mathrm{SU}(3) \to S^5$, together with $\pi_4(S^5) = \pi_3(S^5) = 0$, shows that

$$\pi_3(\mathrm{SU}(3)) \cong \pi_3(\mathrm{SU}(2)) = \pi_3(S^3) \cong \mathbb{Z}.$$

Choose the generator so that its associated bundle has second Chern number $+1$. A clutching class $k \in \mathbb{Z}$ then has second Chern number k. Extending the explicit product framing in theorem 1.3 by the same framing gives the constant clutching map and $k = 0$. Neither $A^\dagger A$ nor $\|a \wedge b\|^2$ selects a nonconstant κ. Thus $k \neq 0$ is valid additional gauge data, but it is not produced by the Gram construction. □

There are two further cohomological checks. If a candidate total space is diffeomorphic to S^6, then

$$H^2(S^6; \mathbb{Z}) = H^4(S^6; \mathbb{Z}) = 0.$$

Consequently no nonzero fibrewise degree-two class can be the restriction of a degree-two class on that total space, and no c_2 on the total space can restrict to a nonzero degree-four class on a fibre. This does not forbid independently chosen higher characteristic data; it proves exactly that the degree-two and degree-four classes in question do not extend.

Likewise, for every continuous map $f : S^{12} \to S^4$ and generator $u_4 \in H^4(S^4;\mathbb{Z})$,

$$f^* u_4 = 0$$

because $H^4(S^{12};\mathbb{Z}) = 0$. Therefore the pullback along any such map of an SU(3) bundle with nonzero $c_2 = ku_4$ has $c_2 = 0$. This statement does not assert that every intrinsic SU(3) bundle on S^{12} is trivial; it says that an S^4 second-Chern charge is not inherited by pullback to the representation sphere.

1.8 The log-Euclidean metric and Yang–Mills theory are extra choices

The logarithm identifies the open positive cone with a real vector space. One may therefore declare

$$X = \log H, \qquad g_{\log} = \frac{1}{2}\operatorname{tr}(dX\, dX), \tag{1.27}$$

which is a complete flat Euclidean metric in the X-coordinates. This metric is not induced by the Frobenius metric on $\mathscr{V}$. To see the difference without discarding any scale factor, set

$$H(s) = sI_2, \qquad A(s) = \sqrt{s}\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad s > 0.$$

For $\xi \in \mathfrak{su}(3)$,

$$\operatorname{Re}\operatorname{tr}(\dot{A}^\dagger \xi A) = \frac{\dot{s}}{2}\operatorname{Re}(\xi_{11} + \xi_{22}) = 0,$$

so $\dot{A}$ is horizontal for the Frobenius quotient. Hence

$$\|\dot{H}\|^2_{\text{quot}} = \|\dot{A}\|^2_F = \frac{\dot{s}^2}{2s}. \tag{1.28}$$

By contrast, $X(s) = (\log s)I_2$, and

$$\|\dot{H}\|^2_{\log} = \frac{1}{2}\operatorname{tr}(\dot{X}^2) = \frac{\dot{s}^2}{s^2}. \tag{1.29}$$

The ratio of (1.28) and (1.29) depends on s, so no global constant rescaling identifies the metrics. The Lorentz form (1.16) is a third object and is indefinite.

After choosing (1.27), one may define classical Yang–Mills theory on the trivial product bundle supplied by theorem 1.3. The additional choice of the zero connection has $F_0 = 0$, so it solves the nonlinear classical equation $d_0^* F_0 = 0$; its Coulomb Hessian is the transverse free Laplacian. These are the standard zero-charge vacuum statements on a chosen Euclidean $\mathbb{R}^4$. The Gram morphism itself supplies no nonzero clutching class, no nonflat connection, no interacting Hamiltonian, no quantum Hilbert-space identification, and no mass-gap estimate. Thus the exact logical implication is

four-dimensional Gram/Lorentz quotient: proved;
$S^{12}_{\text{col}}/\mathrm{SU}(3) \cong B^4$: proved;
nonzero native SU(3) sector on S^4 : not produced;
interacting or quantum Yang–Mills inheritance: not produced.

2 The marked rank-six toroidal lattice and the Weyl carrier

The rank-six lattice in this section is the first homology of the marked rank-three torus. It is not the rank-six degree-four defect group introduced in section 2.5 below. Put

$$L_3 = \mathbb{Z}^3, \qquad \bar{L}_3 = \mathbb{Z}^3, \qquad \Lambda_3 = L_3 \oplus \bar{L}_3 = H_1(T^6;\mathbb{Z}), \qquad u = (1,1,1)^{\mathsf{T}}.$$

For $\sigma \in S_3$, let $R_\sigma e_i = e_{\sigma(i)}$. The retained Weyl action on Λ_3 is the diagonal action

$$\sigma(a,b) = (R_\sigma a, R_\sigma b). \tag{2.1}$$

The monodromy, nilpotent part, filling map, and integral symplectic form are

$$M_3(a,b) = (a+b,b), \qquad N_3 = M_3 - I, \qquad c_3(a,b) = b, \tag{2.2}$$

$$\Omega_3((a,b),(a',b')) = a^{\mathsf{T}}b' - b^{\mathsf{T}}a'. \tag{2.3}$$

Proposition 2.1 (Integral Weyl and monodromy structure)**.** *The S_3-action commutes with M_3, preserves c_3 equivariantly, and preserves Ω_3. Moreover*

$$N_3^2 = 0, \qquad \ker N_3 = \operatorname{im} N_3 = L_3 \oplus 0.$$

As an integral S_3-module,

$$\Lambda_3 \cong \mathbb{Z}^3_{\mathrm{perm}} \oplus \mathbb{Z}^3_{\mathrm{perm}};$$

over $\mathbb{Q}$ this is

$$\Lambda_3 \otimes \mathbb{Q} \cong 2\,\mathbf{1} \oplus 2\,\mathrm{Std}. \tag{2.4}$$

Its character on the conjugacy classes represented by $1, (12), (123)$ is respectively $(6,2,0)$.

Proof. Every permutation matrix is orthogonal and fixes u. Substitution in (2.2) shows that the diagonal action commutes with the shear and that $c_3(\sigma(a,b)) = R_\sigma c_3(a,b)$. Orthogonality gives

$$\Omega_3(\sigma(a,b),\sigma(a',b')) = a^{\mathsf{T}}R_\sigma^{\mathsf{T}}R_\sigma b' - b^{\mathsf{T}}R_\sigma^{\mathsf{T}}R_\sigma a' = \Omega_3((a,b),(a',b')).$$

Also $N_3(a,b) = (b,0)$, proving the two nilpotent assertions. Each of the two displayed bases is a three-point permutation orbit. A permutation module $\mathbb{Q}^3$ is $\mathbf{1} \oplus \mathrm{Std}$; doubling gives (2.4). The traces are twice the numbers of fixed points of a permutation acting on three letters, hence $6, 2, 0$. □

2.1 The exact quotient and its nonsplitting as a Weyl lattice

Let $A_F = B_F = \mathbb{Z}^2$, and retain the matrices

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad Q = B_0^{-1}P = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix}. \tag{2.5}$$

Define

$$\Phi = P \oplus Q : \Lambda_3 \longrightarrow A_F \oplus B_F, \qquad \Phi(a,b) = (Pa, Qb), \tag{2.6}$$

and

$$M_F(x,y) = (x + B_0 y, y), \qquad c_F(x,y) = y.$$

Theorem 2.2 (Kernel, weight lattices, and integral nonsplitting)**.** *The map Φ is surjective and fits into an exact sequence of integral S_3-modules*

$$0 \longrightarrow K \longrightarrow \Lambda_3 \xrightarrow{\Phi} A_F \oplus B_F \longrightarrow 0, \qquad K = \mathbb{Z}(u,0) \oplus \mathbb{Z}(0,u). \tag{2.7}$$

The action on A_F and B_F is the uniquely induced action satisfying

$$\rho_A(\sigma)P = PR_\sigma, qquad\rho_B(\sigma)Q = QR_\sigma. \tag{2.8}$$

It obeys

$$\rho_A(\sigma)B_0 = B_0\rho_B(\sigma), qquad\Phi M_3 = M_F\Phi, qquad c_F\Phi = Qc_3.$$

Both A_F and B_F, with these actions, are the A_2 weight lattice $P(A_2) = \mathbb{Z}^3/\mathbb{Z}u$. They are not the root lattice

$$Q(A_2) = \{a \in \mathbb{Z}^3 : u^{\mathsf{T}}a = 0\},$$

whose image has index three in either quotient. Sequence (2.7) splits in the category of abelian groups, but it has no integral S_3-equivariant splitting.

Proof. The two-by-two minors of P include 1, so P is surjective; the identity $Q = B_0^{-1}P$, with $B_0 \in \mathrm{SL}_2(\mathbb{Z})$, gives the same conclusion for Q. Direct solution of $Pa = 0$ gives $a_1 = a_2 = a_3$, and therefore $\ker P = \ker Q = \mathbb{Z}u$. This proves (2.7), and K is fixed pointwise by S_3.

Because P and Q are quotient maps with invariant kernel, the formulas (2.8) define unique integral actions. From $B_0Q = P$ and the surjectivity of Q,

$$\rho_A(\sigma)B_0Q = \rho_A(\sigma)P = PR_\sigma = B_0QR_\sigma = B_0\rho_B(\sigma)Q,$$

which proves the first compatibility. The other two are the direct calculations

$$\Phi M_3(a,b) = (Pa + Pb, Qb) = (Pa + B_0Qb, Qb) = M_F\Phi(a,b)$$

and $c_F\Phi(a,b) = Qb = Qc_3(a,b)$.

The first isomorphism theorem identifies each quotient with $\mathbb{Z}^3/\mathbb{Z}u$. In the root basis $\alpha_1 = e_1 - e_2, \alpha_2 = e_2 - e_3$,

$$P(\alpha_1\ \alpha_2) = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}. \tag{2.9}$$

The determinant is three, proving that the root lattice maps with index three into the weight lattice. An explicit abelian-group section of P is

$$S = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & -1 \end{pmatrix}, \qquad PS = I_2;$$

an explicit section of Q is SB_0. Hence $(x,y) \mapsto (Sx, SB_0y)$ splits Φ as abelian groups.

It remains to prove the integral equivariant obstruction. Over $\mathbb{Q}$ the invariant line $\mathbb{Q}u$ has the unique isotypic complement $u^\perp = \{a : u^{\mathsf{T}}a = 0\}$; there are no equivariant maps from Std to **1**. Thus any rational S_3-equivariant complement to $K \otimes \mathbb{Q}$ in $\Lambda_3 \otimes \mathbb{Q}$ must be $u^\perp \oplus u^\perp$. Its integral intersection is $Q(A_2) \oplus Q(A_2)$, whose image in $A_F \oplus B_F$ has index $3^2 = 9$ by (2.9). It is not surjective, so an integral equivariant section cannot exist. □

Proposition 2.3 (The symplectic position of the kernel)**.** *In the ordered basis* $((u,0),(0,u))$*,*

$$\Omega_3|_K = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}. \tag{2.10}$$

Moreover

$$K^{\perp_{\Omega_3}} = Q(A_2) \oplus Q(A_2), \qquad [\Lambda_3 : K \oplus K^{\perp_{\Omega_3}}] = 9.$$

In the root bases (α_1, α_2) *in the two summands, the restricted form is*

$$\Omega_3|_{K^{\perp_{\Omega_3}}} = \begin{pmatrix} 0 & C \\ -C & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}. \tag{2.11}$$

Both K *and its symplectic orthogonal are preserved by* S_3 *and* M_3*. In particular,* Ω_3 *does not descend through* Φ *to a form on* $A_F \oplus B_F$*.*

Proof. Equation (2.10) follows from $u^{\mathsf{T}}u = 3$. Orthogonality to $(u,0)$ and $(0,u)$ is respectively the pair of conditions $u^{\mathsf{T}}b = 0$ and $u^{\mathsf{T}}a = 0$. The determinant of the three-column matrix $(u\ \alpha_1\ \alpha_2)$ has absolute value three; taking both summands gives index nine. Finally $(\alpha_i^{\mathsf{T}}\alpha_j)_{ij} = C$, which proves (2.11). The preservation statements follow from theorem 2.1. A bilinear form descending through a quotient must annihilate the kernel in each argument, whereas (2.10) is nonzero. □

The simultaneous quotient–kernel matrices

$$A = \begin{pmatrix} P \\ e_3^{\mathsf{T}} \end{pmatrix}, \qquad B = \begin{pmatrix} Q \\ e_3^{\mathsf{T}} \end{pmatrix}, \qquad J = \operatorname{diag}(A, B)$$

are unimodular and split the complex-torus lattice and its monodromy. They do not preserve the particular marked alternating form (2.3). In fact, with $\Omega_3 = \left(\begin{smallmatrix} 0 & I_3 \\ -I_3 & 0 \end{smallmatrix}\right)$,

$$J^{\mathsf{T}}\Omega_3 J = \begin{pmatrix} 0 & H \\ -H^{\mathsf{T}} & 0 \end{pmatrix}, \qquad H = A^{\mathsf{T}}B = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 1 \end{pmatrix} \neq I_3. \tag{2.12}$$

Thus this product splitting is an unpolarized complex-torus isomorphism; it cannot be called an isomorphism of principally polarized tori for the standard marked form without an additional polarization comparison.

2.2 Six chambers, six formal rays, and their exact incidence

Let E_{ii} be the diagonal matrix unit and put

$$\nu_i = E_{ii} - \frac{1}{3}I_3.$$

For a one-line permutation $\pi = (\pi_1\pi_2\pi_3) \in S_3$, the marked A_2 chamber is

$$\mathcal{C}_\pi = \mathbb{R}_{\geq 0}\nu_{\pi_1} + \mathbb{R}_{\geq 0}(-\nu_{\pi_3}). \tag{2.13}$$

These are the images under the angular quotient P of the six Kuhn tetrahedra incident to a marked vertex. Notice that both rays in (2.13) are angular: the second is the image of a negative angular vector, not of a deck vector under Q.

Let

$$\mathscr{C} = \mathbb{Z}\langle c_\pi : \pi \in S_3 \rangle \cong \mathbb{Z}[S_3]$$

be the chamber module and let

$$\mathscr{R} = \mathbb{Z}\langle r_1^+, r_2^+, r_3^+, r_1^-, r_2^-, r_3^- \rangle$$

be the formal oriented-ray module. The action permutes indices. Define

$$I : \mathscr{C} \longrightarrow \mathscr{R}, \qquad I(c_\pi) = r_{\pi_1}^+ + r_{\pi_3}^-, \tag{2.14}$$

$$d : \mathscr{R} \longrightarrow \mathbb{Z}, \qquad d(r_i^+) = 1, \qquad d(r_i^-) = -1. \tag{2.15}$$

Theorem 2.4 (Integral chamber–ray incidence)**.** *There is an exact sequence of integral S_3-modules*

$$0 \longrightarrow \mathbb{Z}_{\mathrm{sgn}} \xrightarrow{1\mapsto\varepsilon} \mathscr{C} \xrightarrow{I} \mathscr{R} \xrightarrow{d} \mathbb{Z}_{\mathrm{triv}} \longrightarrow 0, \tag{2.16}$$

where

$$\varepsilon = \sum_{\pi \in S_3} \mathrm{sgn}(\pi) c_\pi. \tag{2.17}$$

Use the chamber order

$$(123), (132), (213), (231), (312), (321)$$

and the ray order

$$(r_1^+, r_2^+, r_3^+, r_1^-, r_2^-, r_3^-).$$

Then the matrix of I is

$$[I] = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}. \tag{2.18}$$

With the basis vectors orthonormal, the chamber incidence pairing is

$$I^{\mathsf{T}} I = \begin{pmatrix} 2 & 1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 1 & 0 \\ 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{pmatrix}, \tag{2.19}$$

and its integral radical is $\mathbb{Z}\varepsilon$.

Proof. The six possible ordered pairs (π_1, π_3) are precisely the pairs (i, j) with $i \neq j$, giving (2.18). After replacing each negative-ray basis vector by its negative, this is the oriented vertex–edge incidence map of the connected bipartite graph having vertices r_i^+, r_j^- and one edge for every $i \neq j$. That graph has six vertices and six edges and is a single six-cycle. Its integral cellular chain complex is exact except for its rank-one first homology. Returning to the original signs identifies the last vertex augmentation with d, and the cycle generator with (2.17). Left multiplication by σ sends ε to $\mathrm{sgn}(\sigma)\varepsilon$, proving the module labels in (2.16). Direct multiplication of (2.18) gives (2.19). Since the Euclidean ray pairing is positive definite, $\ker(I^{\mathsf{T}} I) = \ker I = \mathbb{Z}\varepsilon$. □

The chamber set is one free S_3-orbit, so its permutation character is $(6,0,0)$. By contrast, the six displayed basis vectors of Λ_3 are two three-point orbits and have character $(6,2,0)$ by theorem 2.1. Thus the six chambers and the six basis directions are not isomorphic as S_3-sets or permutation modules. There is an abstract S_3-module isomorphism

$$\psi : \Lambda_3 \xrightarrow{\sim} \mathscr{R}, qquad \psi(e_i,0) = r_i^+, quad \psi(0,e_i) = r_i^-, \tag{2.20}$$

but it is not induced by the fan: both geometric ray signs in (2.13) come from L_3, while $\bar{L}_3$ records deck and filling data.

2.3 Spectral-floor evaluation and the monodromy obstruction

Let

$$\mathscr{D}_{\mathbb{Z}} = \bigoplus_{i=1}^{3} \mathbb{Z}E_{ii} \subset \mathrm{Herm}_3(\mathbb{C}), \qquad F(H) = H - \lambda_{\min}(H) I_3.$$

The diagonal Weyl action is conjugation by permutation matrices. Since

$$F(\nu_i) = E_{ii}, \qquad F(-\nu_i) = I_3 - E_{ii},$$

the endpoint values define the unique S_3-equivariant homomorphism

$$E : \mathscr{R} \longrightarrow \mathscr{D}_{\mathbb{Z}}, qquad E(r_i^+) = E_{ii}, \quad E(r_i^-) = I_3 - E_{ii}. \tag{2.21}$$

Proposition 2.5 (Exact diagonal Gram-boundary carrier)**.** *The map E is surjective and*

$$\ker E = \mathbb{Z} \left\langle \begin{array}{l} r_1^- - r_2^+ - r_3^+, \\ r_2^- - r_1^+ - r_3^+, \\ r_3^- - r_1^+ - r_2^+ \end{array} \right\rangle. \tag{2.22}$$

For the trace pairing on diagonal matrices, the pullback matrix on $\mathscr{R}$ is

$$E^{\mathsf{T}} E = \begin{pmatrix} I_3 & J_3 - I_3 \\ J_3 - I_3 & I_3 + J_3 \end{pmatrix}, \tag{2.23}$$

where J_3 is the all-ones matrix; its radical is $\ker E$. For every chamber,

$$G_\pi := (EI)(c_\pi) = 2E_{\pi_1\pi_1} + E_{\pi_2\pi_2} = F(\nu_{\pi_1} - \nu_{\pi_3}). \tag{2.24}$$

The six G_π are distinct rank-two positive-semidefinite matrices and form a free S_3-orbit on the diagonal part of the cubic Gram boundary. Their integral span is

$$\{\mathrm{diag}(m_1,m_2,m_3) : m_1 + m_2 + m_3 \equiv 0 \pmod 3\}, \tag{2.25}$$

an index-three sublattice of $\mathscr{D}_{\mathbb{Z}}$.

Proof. The positive-ray images already form the standard basis, so E is surjective. In the displayed ray order its matrix is $[I_3 \mid J_3 - I_3]$. Solving its three coordinate equations gives exactly the three independent generators in (2.22); multiplying the matrix by its transpose gives (2.23). On the chamber $s\nu_i + t(-\nu_j)$, where $i \neq j$ and $s,t \geq 0$, the j-th diagonal entry is least, and direct subtraction shows

$$F(s\nu_i - t\nu_j) = sE_{ii} + t(I_3 - E_{jj}).$$

Putting $s = t = 1$ proves (2.24); the ordered diagonal entries are a permutation of $(2,1,0)$, which proves rank, positivity, distinctness, and freeness. Every such vector has coordinate sum three. Conversely, the three columns corresponding to $G_{123}, G_{132}, G_{213}$ have determinant -3. Their span therefore has index three and lies in the index-three lattice on the right of (2.25); equality follows. □

The preceding proposition is a typed map from the chamber/ray module to a rank-three diagonal lattice in the nine-real-dimensional Hermitian carrier. It does not yet give a map from all of Λ_3, and the missing compatibility cannot be repaired by assigning the deck basis to the negative rays. Indeed, the homomorphism $g(e_i) = E_{ii}$ used below is the unique linear extension of the three positive endpoint labels to L_3; it is not a map on the quotient $A_F = L_3/\mathbb{Z}u$, since $g(u) = I_3 \neq 0$. The genuine spectral-floor map is only piecewise linear on the chamber fan.

Theorem 2.6 (No monodromy-compatible extension of the angular Gram rays)**.** *Let*

$$g : L_3 \xrightarrow{\sim} \mathscr{D}_{\mathbb{Z}}, qquad g(a) = \mathrm{diag}(a_1, a_2, a_3).$$

There do not exist homomorphisms $\eta : \Lambda_3 \to \mathscr{D}_{\mathbb{Z}}$ and $T : \mathscr{D}_{\mathbb{Z}} \to \mathscr{D}_{\mathbb{Z}}$ such that

$$\eta|_{L_3 \oplus 0} = g, qquad \eta M_3 = T\eta. \tag{2.26}$$

Consequently no torus homomorphism whose angular cocharacter map is the Gram-ray marking can intertwine the displayed monodromy with an endomorphism of this diagonal Gram carrier.

Proof. For every $a \in L_3$, $M_3(a, 0) = (a, 0)$. Hence (2.26) gives $Tg(a) = g(a)$. Since g is surjective, $T = I$. For every $b \in \bar{L}_3$, the same equation applied to $(0, b)$ now gives

$$g(b) + \eta(0, b) = \eta(b, b) = \eta M_3(0, b) = T\eta(0, b) = \eta(0, b),$$

so $g(b) = 0$ for all b, contradicting that g is an isomorphism. A torus homomorphism induces a homomorphism of cocharacter lattices, and intertwining family monodromies induces the second equality in (2.26); therefore the lattice contradiction is also the stated torus-level obstruction. □

For completeness, the tempting formal assignment is

$$\eta_0 = E\psi, qquad \eta_0(a, b) = \mathrm{diag}(a) + (u^{\mathsf{T}} b) I_3 - \mathrm{diag}(b). \tag{2.27}$$

Its kernel has the basis

$$k_i = (e_i - u, e_i), \qquad i = 1, 2, 3. \tag{2.28}$$

This kernel is a rank-three Lagrangian for Ω_3, because direct substitution gives $\Omega_3(k_i, k_j) = 0$. It is not monodromy invariant:

$$\eta_0(k_i) = 0, qquad \eta_0(M_3 k_i) = E_{ii}. \tag{2.29}$$

Thus (2.29) exhibits the obstruction of theorem 2.6 on three primitive generators.

2.4 The branch map is angular: the corrected β square

Consider the monomial torus map

$$q : (\mathbb{C}^*)^4 \longrightarrow (\mathbb{C}^*)^3, qquad q(z_0, z_1, z_2, z_3) = (z_0 z_3, z_1, z_2).$$

On cocharacter lattices it induces

$$\beta : \mathbb{Z}^4 \longrightarrow \mathbb{Z}^3, qquad \beta(n_0, n_1, n_2, n_3) = (n_0 + n_3, n_1, n_2). \tag{2.30}$$

If Σ_r denotes coordinate sum on $\mathbb{Z}^r$, then

$$\Sigma_3 \beta = \Sigma_4. \tag{2.31}$$

Thus β restricts from the four-branch angular fibre lattice $\ker \Sigma_4$ to the three-branch angular fibre lattice $\ker \Sigma_3$.

Define the integral isomorphisms

$$\alpha_3 : L_3 \xrightarrow{\sim} \ker \Sigma_4, \qquad \alpha_3(x_1, x_2, x_3) = (-x_1, x_1 - x_2, x_2 - x_3, x_3), \tag{2.32}$$

$$\alpha_2 : A_F \xrightarrow{\sim} \ker \Sigma_3, \qquad \alpha_2(y_1, y_2) = (-y_1 - y_2, y_1, y_2). \tag{2.33}$$

Proposition 2.7 (Typed angular factorization)**.** *The exact branch/fan square is*

$$\boxed{\beta\alpha_3 = \alpha_2 P}, \tag{2.34}$$

or equivalently

$$\begin{array}{ccc} L_3 & \xrightarrow{P} & A_F \\ {\scriptstyle \alpha_3}\downarrow{\scriptstyle \cong} & & \downarrow{\scriptstyle \alpha_2} \\ \ker \Sigma_4 & \xrightarrow{\beta|_{\ker \Sigma_4}} & \ker \Sigma_3. \end{array}$$

The deck map $Q : \bar{L}_3 \to B_F$ belongs instead to the separate filling square $c_F \Phi = Q c_3$. Therefore the written matrix product $Q\beta$, although numerically formable after erasing all lattice labels, is not a composition of the displayed geometric maps: the codomain of β is the branch cocharacter lattice $\mathbb{Z}^3$, whereas the domain of Q is the independently marked deck lattice $\bar{L}_3$.

Proof. The matrices of the two vertical maps in the displayed bases are

$$[\alpha_3] = \begin{pmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad [\alpha_2] = \begin{pmatrix} -1 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

They have images in the stated sum-zero lattices, and solving successively for the partial sums gives integral inverses. Multiplication with (2.5) and (2.30) gives

$$[\beta][\alpha_3] = \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix} = [\alpha_2]P,$$

which proves the square. The source and target labels in (2.6) show that Q acts on $\bar{L}_3$, not on the branch coordinate lattice appearing in (2.30); this proves the type statement. The valid deck/filling identity was already proved in theorem 2.2. □

2.5 The distinct regular-locus defect lattice

For the candidate six-dimensional regular locus $U = X \setminus (W \sqcup S_1 \sqcup S_2)$, the independent defect calculation has the marked decomposition

$$\mathcal{L}_{\mathrm{def}} := H^4(U; \mathbb{Z}) \cong H_1(W; \mathbb{Z}) \oplus H_1(S_1; \mathbb{Z}) \oplus H_1(S_2; \mathbb{Z}) \cong (\mathbb{Z}^2)^3. \tag{2.35}$$

The cusp, order-three, and order-four support components distinguish the three summands; no geometric S_3-action permuting them is supplied. In the marked basis $(a_0, b_0, a_1, b_1, a_2, b_2)$, its primitive fibre-degree map is

$$\deg_F : \mathcal{L}_{\mathrm{def}} \longrightarrow \mathbb{Z}, \qquad [\deg_F] = (12, 0, 4, 0, -3, 0). \tag{2.36}$$

Its reductions define finite character quotients such as $\mathcal{L}_{\mathrm{def}}/\ker(\rho_6 \deg_F) \cong \mathbb{Z}/6$, but this is secondary quotient data of $\mathcal{L}_{\mathrm{def}}$, not its underlying rank-six structure.

There is consequently no typed identification in the retained data between $\mathcal{L}_{\mathrm{def}}$ and Λ_3. Equality of ranks supplies no map; their markings, functorial origins, distinguished maps, and available pairings differ. In particular Ω_3 is the explicitly chosen nondegenerate alternating form in the marked decomposition $L_3 \oplus \bar{L}_3$, not an intersection pairing canonically attached to the six-torus. Ordinary Poincaré duality for the noncompact six-dimensional regular locus pairs $H^4(U)$ with compactly supported degree-two data and does not provide the displayed self-pairing on $\mathcal{L}_{\mathrm{def}}$. A comparison would require an explicit geometric correspondence

$$H_1(T^6;\mathbb{Z}) \longrightarrow H^4(U;\mathbb{Z})$$

together with proofs of its compatibility with monodromy, support markings, and the relevant pairings. No such correspondence occurs in the audited construction.

2.6 What the exact carrier does and does not supply

The verified maps now form three distinct diagrams:

$$\Lambda_3 \xrightarrow{P\oplus Q} A_F \oplus B_F, \qquad \mathscr{C} \xrightarrow{I} \mathscr{R} \xrightarrow{E} \mathscr{D}_{\mathbb{Z}}, \qquad \ker \Sigma_4 \xrightarrow{\beta} \ker \Sigma_3.$$

The first intertwines monodromy and filling, the second records angular Weyl-chamber incidence and its diagonal spectral-floor image, and the third is the cocharacter map of the local monomial toric morphism. The equality of several ranks or the possibility of multiplying their unlabeled integer matrices does not merge these diagrams. In particular, the construction provides six exact marked points in the rank-two stratum of the cubic Gram boundary, but it does not provide a continuous, holomorphic, toric, or stratified map from the compactified toroidal total space to the full eight-dimensional Gram boundary. Such a global map would still require explicit overlap descent, a singular-fibre extension, and a boundary comparison compatible with monodromy and filling; the lattice obstruction of theorem 2.6 rules out the most direct angular Gram-ray extension.

Conclusion

Two complex color triplets produce an exact four-real-dimensional Gram orbit space, and the compact representation quotient is the four-ball with the complete orbit-type boundary displayed above. Independently, the toroidal homology lattice Λ_3 carries an exact diagonal Weyl action, symplectic shear, and weight-lattice quotient. Its six Kuhn chambers map by an integral incidence complex to six diagonal rank-two points of the cubic Gram boundary. They do not identify with the six basis directions of Λ_3, and no lattice map extending the angular Gram-ray marking can intertwine the toroidal monodromy. The branch map factors through P, not through the deck map Q, and the separately arising free defect lattice $H^4(U;\mathbb{Z})$ remains nonidentified with Λ_3.

Thus the surviving result is an exact collection of typed orbit, lattice, fan, and boundary-carrier theorems together with a sharp monodromy obstruction. It does not supply the missing global toroidal-to-Gram map, a nonzero S^4 clutching class, or a nontrivial interacting Yang–Mills sector.